WorMI: World Model Implanting for Test-time Adaptation of Embodied Agents

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Abstract

Embodied agents driven by large language models often degrade sharply when instructions must be executed in unseen scenes or previously unseen domains, and adapting them by full retraining is costly. We present WorMI, a plug-and-play test-time adaptation framework that composes a frozen reasoning model with a retrieved set of domain-specific world models. WorMIfirst retrieves relevant world models using prototype summaries of textual environment trajectories and Wasserstein-distance matching, then injects their hidden states into the reasoning model with a hierarchical compound attention module that performs world-to-world fusion followed by world-to-reasoning alignment. Only the lightweight composer is trained, while the reasoning model and the world models remain frozen; the composer is optimized with first-order meta-learning across subsets of world models to improve rapid specialization. On VirtualHome and ALFWorld, WorMIconsistently outperforms prior LLM planning baselines in both zero-shot and few-shot transfer. In the hardest zero-shot split with unseen tasks and unseen scenes, WorMIimproves success rate from 45.71% to 66.12% on VirtualHome and from 39.66% to 51.67% on ALFWorld, while also reducing plan steps. Additional analysis shows that prototype-based retrieval, compound attention, and selective model implantation are all important to robust adaptation.

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1. Introduction

Large language models (LLMs) have become strong high-level planners for embodied control, enabling agents to convert natural-language instructions into executable action sequences (Huang et al., 2022a; Brohan et al., 2023; Song et al., 2023; Huang et al., 2022b; Yao et al., 2022). Yet their behavior remains brittle under domain shift. A policy that succeeds in familiar rooms or task distributions often deteriorates when scene layouts, object configurations, or task compositions change, even when the required skills remain conceptually similar. This gap is especially problematic for embodied agents, where failures accumulate over long horizons and on-the-fly full-model retraining is typically infeasible.

Existing approaches usually resolve this tension in one of two ways. The first is to keep the LLM fixed and rely on prompting or in-context examples (Huang et al., 2022a; Song et al., 2023; Yao et al., 2022; Wang et al., 2023), which is lightweight but often insufficient when the target environment requires structured domain knowledge absent from the prompt. The second is to adapt or ground the planner through task-specific modules or finetuning (Brohan et al., 2023; Hazra et al., 2024; Singh et al., 2023; Liang et al., 2023), which can improve performance but weakens modularity and raises deployment cost. More broadly, recent work on model composition shows that pretrained models can be fused or ensembled to transfer complementary capabilities (Wan et al., Jiang et al., 2023). For embodied agents, however, it remains unclear how to compose domain-specific world knowledge at test time without retraining the entire planner.

We address this problem with *world model implanting*: instead of retraining the planner, we maintain a bank of independently trained domain-specific world models and dynamically compose them with a frozen reasoning model at inference time. Our method, WorMI, follows a simple principle. First, select only the world models that are relevant to the current environment. Second, fuse them in a way that preserves domain-specific structure while aligning the result to the reasoning model's hidden states. This yields a modular adaptation mechanism that is lightweight at deployment and naturally supports continual addition or removal of domain knowledge.

WorMIhas two main components. The first is a prototype-based retrieval module that summarizes each world model's training domain using clustered sentence embeddings of textual trajectories and retrieves the nearest models using Wasserstein distance. The second is a world-wise compound attention composer that receives hidden states from the retrieved world models, performs world-level fusion, and then injects the fused representation into selected layers of the frozen reasoning model. Only the composer parameters are trained. To improve generalization to unseen combinations of domains, we optimize the composer with a first-order

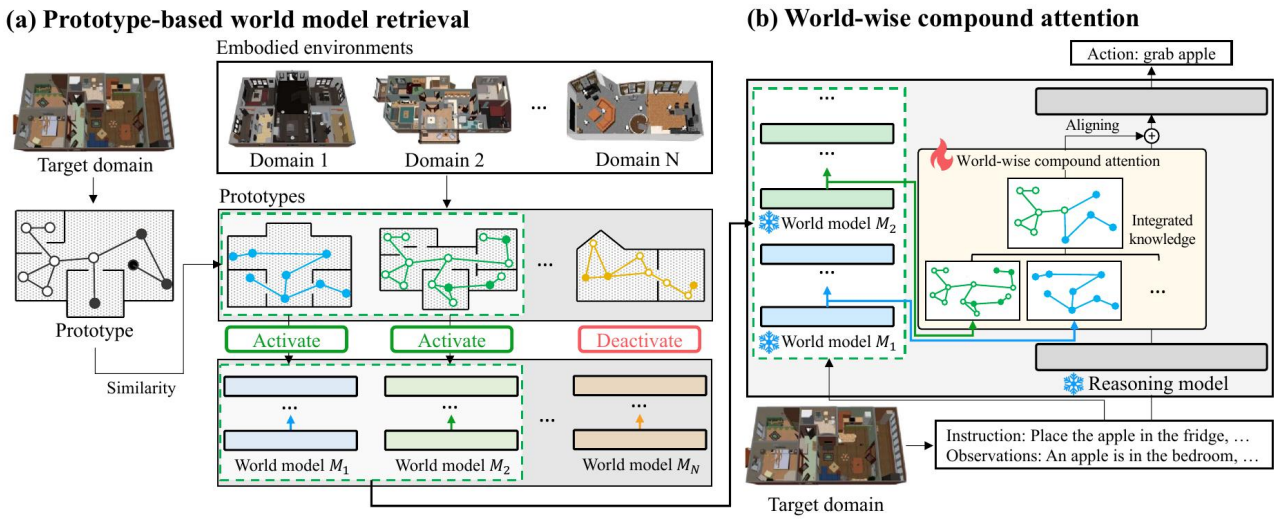

Figure 1: WorMIoverview. Left: prototype-based retrieval activates a small subset of relevant world models from a larger bank. Right: the retrieved world models are implanted into a frozen reasoning model through a world-wise compound attention module, allowing domain-specific knowledge to be added or removed at test time.

meta-learning procedure in the spirit of REPTILE (Nichol, 2018), using curricula defined over subsets of world models.

Experiments on VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021) show that this simple design yields strong gains across zero-shot, few-shot, and complex-instruction settings. The largest improvements occur precisely in the cross-domain regime that motivates the method: on unseen tasks and unseen scenes, WorMIimproves success rate by 20.41 points on VirtualHome and 12.01 points on ALFWorld relative to SayCanPay (Hazra et al., 2024). We further show that retrieval quality matters, compound attention matters, and adding irrelevant or overly many world models can hurt, which justifies explicit selection and compositional control.

Our contributions are:

- We introduce world model implanting, a test-time adaptation framework that composes retrieved domain-specific world models with a frozen LLM reasoner.
- We propose a prototype-based retrieval mechanism and a hierarchical compound attention composer for world-model fusion and reasoning alignment.
- We show strong zero-shot and few-shot transfer on VirtualHome and ALFWorld, together with analyses of retrieval, attention design, world-model scale, and continual implantation behavior.

2. Related Work

2.1. LLM Planning for Embodied Agents

LLMs have been used as zero-shot or few-shot planners for embodied control by translating instructions into action sequences (Huang et al., 2022a; Song et al., 2023; Yao et al., 2022; Huang et al., 2022b). Subsequent systems improved grounding with affordances, search, or structured prompts, including SayCan (Brohan et al., 2023), SayCanPay (Hazra et al., 2024), ProgPrompt (Singh et al., 2023), Code as Policies (Liang et al., 2023), and DEPS (Wang et al., 2023). These methods show that pretrained language models contain useful procedural knowledge, but they still struggle when the target environment requires domain knowledge not well captured by the base model or the prompt. WorMIis complementary: instead of replacing LLM planning, it augments a frozen planner with retrieved world models specialized to different domains.

2.2. Model Composition and Knowledge Fusion

Recent work studies how pretrained models can be combined without full joint retraining. Knowledge fusion and ensembling methods aggregate complementary model outputs or capabilities (Wan et al.; Jiang et al., 2023). These methods are promising for modular adaptation, but they are not designed for embodied planning with dynamic domain selection. WorMIadopts the same modular perspective while introducing two ingredients needed for embodied agents: retrieval over a bank of domain-specific world models, and layer-wise fusion of retrieved world states with a reasoning model.

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2.3. Generalization, Continual Learning, and Meta-Learning

Robustness to distribution shift is a central goal in domain generalization (Zhou et al., 2022). In embodied learning, continual adaptation across tasks and environments remains difficult (Kim et al., 2024). Meta-learning methods such as MAML and REPTILE learn parameters that adapt quickly across task families (Finn et al., 2017; Nichol, 2018). WorMIadopts this intuition at the composer level: rather than adapting the full policy, it meta-trains a lightweight interface across subsets of world models so that new domain combinations can be handled by retrieval and composition at test time.

3. Method

3.1. Problem Setting

We assume a frozen reasoning model π that maps an instruction-observation history to the next action token sequence. In addition, we maintain a bank of independently trained domain-specific world models $\{\mathcal{M}_1, \ldots, \mathcal{M}_N\}$. Each world model is trained on data from a particular source domain and remains frozen after pretraining. At test time, given a target environment and its partial interaction context, our goal is to select a small subset of relevant world models and compose them with π without updating the reasoning model itself.

This setting targets the practically important case where domain knowledge accumulates over time in modular form. A newly trained world model can be added to the bank, and a previously used one can be removed, without retraining the base planner.

3.2. Prototype-Based World Model Retrieval

Using all world models simultaneously is inefficient and, as we show later, often harmful. WorMItherefore retrieves only a small set of relevant world models.

For each source domain corresponding to world model $\mathcal{M}_i$, we encode textual environment trajectories with a frozen sentence encoder Φ and obtain embeddings $\{\mathbf{e}_{i,j}\}_j$. We summarize these embeddings by clustering them into a prototype set $\mathbf{P}_i$. In the released implementation, prototypes are computed by `KMeans` over the sentence embeddings. Given a target interaction context, we build a query prototype set $\mathbf{P}_q$ in the same embedding space and score each world model by Wasserstein distance:

$$i^\star = \arg\min_i \; W(\mathbf{P}_q, \mathbf{P}_i). \tag{1}$$

We retrieve the top-K nearest world models. This retrieval stage makes the framework scalable: the bank can grow, while only a small number of world models are activated for any single episode.

3.3. World Model Implanting

After retrieval, WorMIimplants the selected world models into intermediate layers of the frozen reasoning model. Concretely, for a set of connection layers, we run the selected world models in parallel and extract hidden states from corresponding layers. These world hidden states are not merged at the parameter level. Instead, they are injected into the reasoning model through learned hook modules, which is why models can be added or removed at test time.

Let $\mathbf{h}_\ell^R \in \mathbb{R}^{B\times T\times d_R}$ be the hidden state of the reasoning model at layer ℓ, and let $\mathbf{h}_\ell^{W_m} \in \mathbb{R}^{B\times T\times d_W}$ denote the hidden state from the m-th retrieved world model. A learned projection g_θ maps each world hidden state into the reasoning-model dimensionality:

$$\tilde{\mathbf{h}}_\ell^{W_m} = g_\theta(\mathbf{h}_\ell^{W_m}). \tag{2}$$

This produces a set of aligned world representations that can be fused by the composer.

3.4. World-Wise Compound Attention

The key design question is how to combine multiple world models before injecting them into the reasoning model. Simple concatenation or addition ignores the fact that different world models may be relevant to different tokens or decision stages. WorMIinstead uses a hierarchical composer, illustrated in Figure 2.

The composer has two stages.

World-level fusion. For each token position, the projected world states are stacked along a world dimension and fused with attention over retrieved models. This stage decides *which world models matter* for the current token and context, producing an integrated world representation $\bar{\mathbf{h}}_\ell^W$.

Reasoning-level alignment. The fused world representation is then consumed by a cross-attention block whose query comes from the reasoning-model hidden state $\mathbf{h}_\ell^R$. This stage decides *how the integrated world knowledge should modify the current reasoning state*. The resulting output is added back to the reasoning hidden state as a residual update:

$$\hat{\mathbf{h}}_\ell^R = \mathbf{h}_\ell^R + C_\theta\Big(\mathbf{h}_\ell^R, \{\tilde{\mathbf{h}}_\ell^{W_m}\}_{m=1}^K\Big). \tag{3}$$

Compared with naive fusion, this hierarchy separates inter-world aggregation from world-to-reasoning alignment. The empirical ablations in Table 4 show that this distinction is important.

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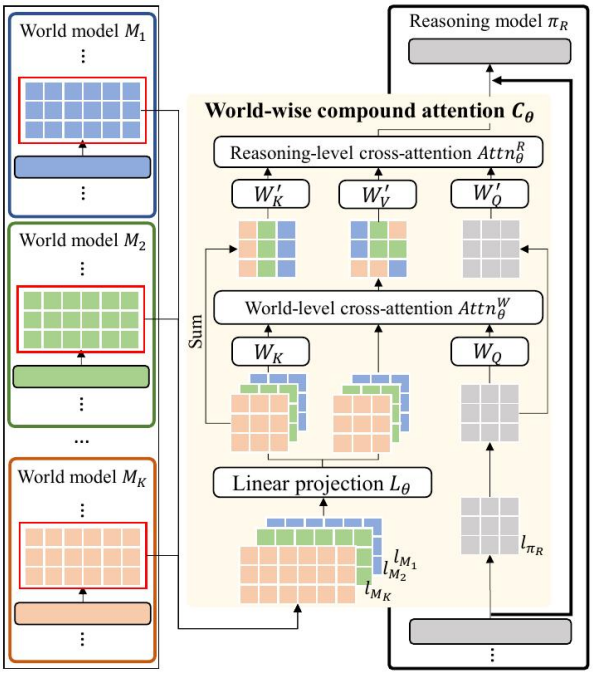

Figure 2: World-wise compound attention. Retrieved world-model hidden states are projected to the reasoning-model space, fused through world-level attention, and then aligned to the frozen reasoning model by reasoning-level cross-attention.

3.5. Training

World models are trained independently on their source-domain data. During WorMItraining, the reasoning model and all world models are frozen, and only the composer parameters θ are optimized. We use supervised autoregressive training on the embodied interaction traces, implemented as chat-formatted action prediction.

To encourage robustness to different combinations of source domains, we train the composer on curricula that each activate a different subset of world models. The implementation uses a first-order meta-learning procedure in the spirit of REPTILE (Nichol, 2018): the composer is trained separately on each curriculum within an iteration, and the resulting parameter sets are averaged to form the next initialization. This procedure trains the composer to work well across world-model subsets rather than overfitting to a single fixed composition.

3.6. Inference

At test time, WorMIfirst retrieves the most relevant world models, implants them into the frozen reasoning model, and then generates actions autoregressively. Because composition happens through hidden-state hooks rather than weight merging, world models can be inserted or removed on demand. This is especially useful in continual settings where the available world-model bank changes over time.

4. Experiments

4.1. Environments, Splits, and Metrics

We evaluate on VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021), two standard benchmarks for embodied instruction following. VirtualHome focuses on household programs executed in simulated homes, while ALFWorld provides text-based embodied tasks aligned with ALFRED-style goals.

We consider zero-shot transfer across three splits: seen tasks with seen scenes, seen tasks with unseen scenes, and unseen tasks with unseen scenes. We also evaluate few-shot adaptation with 1-shot, 5-shot, and 10-shot supervision, plus additional complex-instruction settings involving long-horizon and multiple instructions. Following the provided tables, we report success rate (SR), plan steps (PS), and, where applicable, goal-condition completion (GC). All main tables report 95% confidence intervals over 5 random seeds.

4.2. Baselines

We compare against representative LLM-based embodied planning baselines. ZSP (Huang et al., 2022a) is a zero-shot language planner. LLM-Planner (Song et al., 2023) is a few-shot grounded planner. SayCanPay (Hazra et al., 2024) augments LLM planning with learnable domain knowledge and heuristic search. We also include a finetuned LLM baseline (LLM-FT / LLM+FT in the provided tables). WorMIuses the same overall planning interface but augments the frozen reasoner with retrieved world models.

4.3. Main Results

Table 1 shows that WorMIconsistently outperforms all baselines in zero-shot transfer on both benchmarks. The most important result is the hardest setting, unseen tasks with unseen scenes, where WorMIreaches 66.12% SR on VirtualHome and 51.67% on ALFWorld. Compared with SayCanPay, this is a gain of 20.41 points and 12.01 points, respectively, while also reducing PS by 3.87 and 3.68.

The gains are not confined to the hardest split. On VirtualHome, WorMIimproves SR from 69.88% to 85.78% on seen tasks and seen scenes, and from 64.74% to 80.26% on seen tasks and unseen scenes. On ALFWorld, the corresponding improvements over SayCanPay are from 40.67% to 62.51% and from 34.10% to 52.67%. These results indicate that the composer is not merely memorizing a special-case transfer regime; it systematically improves planning quality across domain conditions.

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Table 1: Zero-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	Seen Tasks & Seen Scenes		Seen Tasks & Unseen Scenes		Unseen Tasks & Unseen Scenes	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome						
ZSP	11.15%±0.65%	29.02±0.08	8.95%±1.13%	29.25±0.11	8.19%±0.33%	29.36±0.56
LLM-FT	58.55%±2.18%	17.37±0.34	53.42%±0.79%	17.70±0.23	42.82%±0.76%	20.79±0.17
LLM-Planner	35.67%±1.25%	27.15±0.13	28.55%±0.42%	27.04±0.12	21.45%±0.42%	27.73±0.05
SayCanPay	69.88%±2.32%	14.53±0.55	64.74%±0.87%	15.64±0.18	45.71%±0.59%	19.04±0.63
WorMI	**85.78%±0.45%**	**10.76±0.19**	**80.26%±1.02%**	**12.42±0.09**	**66.12%±0.80%**	**15.17±0.08**
Evaluation in ALFWorld						
ZSP	2.30%±0.21%	49.34±0.66	2.26%±0.09%	49.04±0.95	2.13%±0.09%	49.68±0.23
LLM+FT	45.72%±0.39%	14.63±1.35	29.26%±0.72%	38.49±2.87	29.40%±1.65%	46.84±0.15
LLM-Planner	17.73%±0.61%	32.09±2.90	18.63%±0.82%	40.50±2.57	12.31%±0.80%	46.33±1.68
SayCanPay	40.67%±1.24%	18.37±1.93	34.10%±1.17%	20.82±1.21	39.66%±1.43%	23.86±1.90
WorMI	**62.51%±1.65%**	**9.96±1.29**	**52.67%±1.39%**	**17.74±0.83**	**51.67%±2.23%**	**20.18±0.63**

Table 2: Few-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	1-Shot		5-Shot		10-Shot		Average	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome								
LLM-FT	42.35%±0.85%	20.46±0.15	47.22%±0.46%	16.82±0.29	51.45%±1.06%	16.82±0.39	47.01%±0.79%	18.03±0.28
LLM-Planner	24.63%±0.34%	26.19±0.31	29.80%±0.55%	26.98±0.08	33.49%±0.44%	26.60±0.16	29.31%±0.44%	26.59±0.18
SayCanPay	46.75%±1.02%	19.12±0.16	49.80%±1.11%	15.52±0.11	56.24%±0.79%	16.00±0.13	50.93%±0.97%	16.88±0.13
WorMI	**74.90%±1.57%**	**12.18±0.31**	**78.04%±0.34%**	**11.85±0.08**	**79.61%±0.99%**	**11.68±0.13**	**77.51%±0.98%**	**11.90±0.17**
Evaluation in ALFWorld								
LLM-FT	26.78%±1.20%	46.51±0.27	29.79%±1.36%	46.21±0.37	33.57%±0.98%	43.70±0.83	30.05%±1.18%	45.47±0.49
LLM-Planner	18.28%±1.06%	43.94±2.87	16.80%±1.46%	40.50±3.14	17.76%±1.25%	42.45±2.64	17.61%±1.26%	42.30±2.88
SayCanPay	40.94%±1.34%	21.58±1.14	36.70%±1.31%	22.15±3.55	39.54%±1.48%	24.74±3.48	39.06%±1.38%	22.82±2.72
WorMI	**51.50%±2.21%**	**21.46±3.96**	**58.69%±1.94%**	**13.14±1.00**	**64.46%±0.88%**	**11.79±0.81**	**58.22%±1.68%**	**15.46±1.92**

Table 2 shows that the same advantage persists in few-shot settings. Averaged across 1, 5, and 10 shots, WorMIimproves over SayCanPay by 26.58 SR points on VirtualHome and 19.16 points on ALFWorld, while also reducing PS by 4.98 and 7.36. The gap widens with more supervision in ALFWorld, where WorMIreaches 64.46% SR at 10-shot, suggesting that the composer can use limited target-domain evidence effectively without updating the frozen reasoning model.

The few-shot results also show that the gains are not coming from longer or more exploratory rollouts. In VirtualHome, WorMIlowers PS relative to SayCanPay at every shot count, from 19.12 to 12.18 in 1-shot and from 16.00 to 11.68 in 10-shot, while SR rises from 46.75% to 74.90% and from 56.24% to 79.61%. ALFWorld exhibits the same pattern more strongly at higher shots: by 10-shot, WorMIimproves SR from 39.54% to 64.46% and nearly halves PS from 24.74 to 11.79, indicating more decisive planning rather than simple trial-and-error.

4.4. Backbone Sensitivity

Table 3 studies the impact of the underlying reasoning-model scale. WorMIimproves over both LLM-Planner and SayCanPay for all three tested Llama-3.2 backbones, including 1B, 3B, and 11B variants. The absolute best result is obtained with the 11B reasoner, but the relative advantage remains visible even for the 1B model. This suggests that world model implanting is not tied to a single backbone size; it acts as a modular upgrade to the planner.

Table 3: Impact of LLMs on WorMI and baselines

LLM	Model	SR (↑)	PS (↓)
Llama-3.2-11B	LLM-Planner	51.11%±0.27%	18.33±0.08
	SayCanPay	49.28%±0.20%	18.88±0.10
	WorMI	**71.18%±0.62%**	**13.95±0.11**
Llama-3.2-3B	LLM-Planner	21.45%±0.42%	27.73±0.05
	SayCanPay	45.71%±0.59%	19.04±0.05
	WorMI	**66.12%±0.80%**	**15.17±0.08**
Llama-3.2-1B	LLM-Planner	9.93%±0.27%	27.37±0.11
	SayCanPay	42.22%±1.32%	21.34±0.33
	WorMI	**49.65%±0.83%**	**18.99±0.30**

Quantitatively, the SR gain over SayCanPay is 21.90 points for the 11B reasoner, 20.41 points for the 3B reasoner, and 7.43 points for the 1B reasoner. The smaller but still positive gain at 1B suggests that implantation cannot fully compensate for an underpowered base planner, but remains useful

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as long as the reasoner can exploit the injected world states.

5. Analysis

5.1. Retrieval and Fusion Ablations

Table 4: Ablation study on prototype-based world model retrieval and world-wise compound attention

(a) Prototype-based world model retrieval

Model	**SR (↑)**	**PS (↓)**
WorMI-E	48.63%±0.25%	18.93±0.49
WorMI-R	62.04%±0.44%	16.96±0.20
WorMI	**66.12%±0.80%**	**15.17±0.08**

(b) World-wise compound attention

Model	**SR (↑)**	**PS (↓)**
WorMI-CONCAT	47.79%±2.39%	20.03±0.07
WorMI-ADD	56.47%±0.61%	17.43±0.30
WorMI	**66.12%±0.80%**	**15.17±0.08**

Table 4 isolates the two core components of WorMI. Replacing prototype-based retrieval with a weaker alternative degrades performance from 66.12% to 62.04% SR, and a further weaker variant drops to 48.63%. Likewise, replacing compound attention with simple addition or concatenation reduces SR to 56.47% and 47.79%, respectively. The same trend is reflected in PS. These results support the central claim of the paper: good adaptation requires both selecting the right world models and fusing them with a token-sensitive hierarchical mechanism.

5.2. How Many World Models Should Be Implanted?

Table 5 shows that more world models are not always better. Performance improves sharply when moving from one to three world models, peaking at 66.12% SR with three models, then degrades when four or six are used. This behavior is consistent with the motivation for retrieval: relevant world knowledge is helpful, but excessive or weakly related world models introduce interference. The result also explains why explicit retrieval is preferable to indiscriminate aggregation over the entire model bank.

5.3. Complex Instructions

Complex instructions amplify compounding errors, making them a useful stress test for compositional adaptation. In Table 6, WorMIimproves long-horizon SR from 6.54% to 19.61% and GC from 39.22% to 49.35% over SayCanPay. For multiple instructions, the gains are similarly large: SR improves from 6.54% to 20.26% and GC from 42.32% to 60.62%. The improvements in PS indicate that the method is not merely reaching partial progress through longer trajectories; it is also producing more efficient plans.

Table 5: Impact of world model scale (WMs)

Num. of WMs	**SR (↑)**	**PS (↓)**
1	42.75%±0.50%	20.54±0.02
2	62.94%±0.12%	17.21±0.20
3	66.12%±0.59%	15.17±0.08
4	61.96%±0.25%	16.68±0.34
6	48.23%±0.25%	18.93±0.49

Table 6: Performance for complex instruction scenarios

Model	**SR (↑)**	**GC (↑)**	**PS (↓)**
Long horizon instructions			
LLM-Planner	0.65%±0.51%	17.16%±0.58%	86.12±1.22
SayCanPay	6.54%±0.51%	39.22%±0.66%	57.87±0.28
WorMI	**19.61%±0.88%**	**49.35%±0.55%**	**53.15±0.70**
Multiple instructions			
LLM-Planner	1.31%±0.51%	29.02%±0.08%	85.17±0.76
SayCanPay	6.54%±0.52%	42.32%±0.41%	57.94±0.92
WorMI	**20.26%±1.34%**	**60.62%±0.89%**	**43.49±1.23**

5.4. Attention Behavior and Continual Implanting

Figure 3 provides a qualitative view of the world-level attention. Different world models dominate different instruction segments, such as room-specific subtasks involving a television, cabinet, sofa, and microwave. This is precisely the behavior the compound attention was designed to induce: instead of collapsing all world models into a static mixture, the composer reweights them over time according to the current subgoal.

Figure 4 further demonstrates the modularity of WorMI. As world models are implanted sequentially, success rate increases steadily. Conversely, removing the world model most relevant to the current domain causes substantial degradation, with drops larger than 20 points in all shown domains and as large as 31.20 points in one case. Together, these results indicate that world models contribute non-redundant domain knowledge and that the implanting mechanism can exploit this knowledge compositionally.

Discussion. Two observations emerge from the analyses. First, composition helps only when selection is disciplined: irrelevant world models can dilute useful signals. Second, the best-performing fusion strategy is the one that explicitly models both inter-world competition and world-to-reasoning alignment. These points explain why WorMIremains effective as a test-time adaptation mechanism rather than degenerating into a generic ensemble.

The results also clarify the operating regime in which implantation is most useful. When transfer is easy, such as seen tasks in seen scenes, the gains are already meaningful but moderate; when the agent must extrapolate to unseen tasks, unseen scenes, or multi-step instructions, the margins

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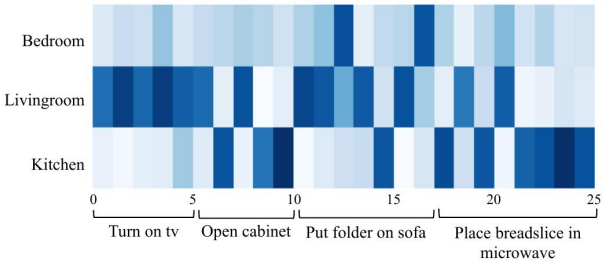
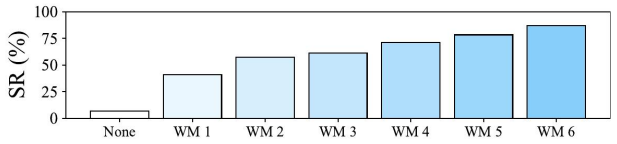
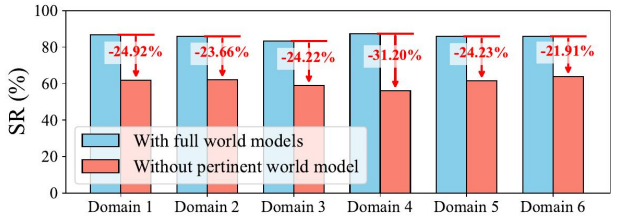


Figure 3: World-level attention map. Attention shifts across room-specific world models as the instruction progresses through subtasks, indicating token-dependent selection of domain knowledge.



(a) Continual implantation from WM1 to WM6.



(b) Removing the pertinent world model causes large drops.

Figure 4: Continual model implanting analysis. Performance improves as relevant world models are added, while removing the pertinent model causes substantial degradation, with annotated SR drops between 21.91 and 31.20 points across domains.

increase substantially. This pattern is consistent with the intended role of world models: they do not replace the base planner's general language competence, but supply missing environment regularities exactly where prompt-only reasoning is least reliable. In that sense, retrieval acts as a guardrail against negative transfer, while compound attention determines when the injected knowledge should influence the decision process.

The ablations also expose a concrete failure mode. When retrieval misses the pertinent source domain, fusion over the remaining models does not reliably recover the missing structure: weak retrieval and naive fusion both fall far below the full system in SR. The peak at three implanted world models further suggests that most episodes benefit from a small set of complementary priors rather than broad coverage from the whole bank. This supports viewing WorMIas a sparse routing mechanism, where retrieval narrows the candidate domains and the composer decides, token by token, whether the injected evidence should refine or override the frozen planner. The continual-implanting curves are consistent with this interpretation: adding relevant models yields gains, whereas removing the pertinent model triggers immediate 21.91–31.20 point SR drops. The model bank therefore acts less like redundant memory and more like a set of specialized experts whose utility depends on routing.

6. Conclusion

We presented WorMI, a test-time adaptation method for embodied agents based on world model implanting. WorMIretrieves a small set of relevant domain-specific world models, fuses them with hierarchical compound attention, and injects the result into a frozen reasoning model without retraining the full planner. Across VirtualHome and ALFWorld, the method improves zero-shot, few-shot, and complex-instruction performance, with the largest margins in cross-domain generalization settings. The ablations and qualitative analyses show that both selective retrieval and hierarchical fusion are necessary for these gains, supporting scalable adaptation by expanding the world-model bank as domains evolve.

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